@ Libs Solutions

Zimsec N2024 P1

Marking Guide

We Sink Black !!!

1.

$$\frac{(x^2)^{\frac{1}{3}} \times (x^{-\frac{1}{2}})^3}{x^{-\frac{1}{6}} \times x^{-\frac{2}{3}}}$$

$$= \frac{x^{2(\frac{1}{3})} \times x^{-\frac{1}{2}(3)}}{x^{-\frac{1}{6}} \times x^{-\frac{2}{3}}}$$

$$= \frac{x^{\frac{2}{3}} \times x^{-\frac{3}{2}}}{x^{-\frac{1}{6}+(-\frac{2}{3})}}$$

$$= \frac{x^{-\frac{5}{6}}}{x^{-\frac{5}{6}}}$$

$$= 1$$

Be Addicted to Progress # Libs

Value of r^4

2. (a)

$$S_\infty = 2S_8$$

$$\frac{a}{1-r} = 2\left(\frac{a(1-r^8)}{1-r}\right)$$

$$\cancel{\frac{a}{1-r}} = 2\left(\cancel{\frac{a}{1-r}}(1-r^8)\right)$$

$$\frac{1}{2} = \frac{2(1-r^8)}{2}$$

$$1-r^8 = \frac{1}{2}$$

$$(r^8)^{\frac{1}{2}} = \left(\frac{1}{2}\right)^{\frac{1}{2}}$$

$$r^4 = \left(\frac{1}{2}\right)^{\frac{1}{2}}$$

$$r^4 = \pm\sqrt{\frac{1}{2}}$$

but r, $0 < r \leq 1$

$$\therefore r^4 = \sqrt{\frac{1}{2}}$$

Exact Value of a

(b) $a \neq 0$

$$U_n = ar^{n-1}$$

$$U_9 = 20$$

$$ar^8 = 20$$

$$r^8 = \frac{1}{2}$$

$$a\left(\frac{1}{2}\right) = 20$$

$$\frac{a}{2} = \frac{40}{1}$$

$$\therefore a = 40$$

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3 (a) $y \propto \frac{\sqrt{x}}{z^3}$

$y = \frac{k\sqrt{x}}{z^3}$

$y = 16$; when $x = 9$, $z = \frac{1}{2}$

$16 = \frac{k\sqrt{9}}{(\frac{1}{2})^3}$

$\frac{16}{1} = \frac{k\sqrt{9}}{\frac{1}{8}}$

$16\left(\frac{1}{8}\right) = 3k$

$\frac{2}{3} = \frac{\cancel{3}k}{\cancel{3}}$

$k = \frac{2}{3}$

$\therefore y = \frac{2\sqrt{x}}{3z^3}$

Find the Value of z

(b.) When $y = 2.5$ and $x = 900$

$$y = \frac{2\sqrt{x}}{3z^3}$$

$$\frac{2.5}{1} = \frac{2\sqrt{900}}{3.z^3}$$

$$3(2.5).z^3 = 2(30)$$

$$\frac{\cancel{3(2.5)}z^3}{\cancel{3(2.5)}} = \frac{60}{3(2.5)}$$

$$z^3 = 8$$

$$\left(z^{\cancel{3}}\right)^{\frac{1}{\cancel{3}}} = (8)^{\frac{1}{3}}$$

$$z = (8)^{\frac{1}{3}}$$

$$\therefore z = \sqrt[3]{8}$$

$$\therefore z = 2$$

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4 $$\frac{3-x+6x^2}{(1-x)(2+x)(1+2x)} = \frac{A}{1-x} + \frac{B}{2+x} + \frac{C}{1+2x}$$

$$3-x+6x^2 = A(2+x)(1+2x) + B(1-x)(1+2x) + C(2+x)(1-x)$$

let $x = 1$

$$3-1+6(1)^2 = A(3)(3)$$

$$\frac{8}{9} = \frac{9A}{9}$$

$$A = 8/9$$

let $x = -2$

$$3-(-2)+6(-2)^2 = B(3)(-3)$$

$$5+24 = -9B$$

$$\frac{29}{-9} = \frac{-9B}{-9}$$

$$B = -\frac{29}{9}$$

let $x = -\frac{1}{2}$

$$3-(-\tfrac{1}{2})+6(-\tfrac{1}{2})^2 = C\left(\frac{3}{2}\right)\left(\frac{3}{2}\right)$$

$$\frac{5}{1} = \frac{9C}{4}$$

$$\frac{20}{9} = \frac{9C}{9}, \quad C = \frac{20}{9}$$

$$\therefore \frac{3-x+6x^2}{(1-x)(2+x)(1+2x)} = \frac{8}{9(1-x)} - \frac{29}{9(2+x)} + \frac{20}{9(1+2x)}$$

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Value of a

5 (a)

$$
\begin{array}{r|l}
 & x^2 + 4x + 3 \\
\hline
x^2 - x + 1 & x^4 + 3x^3 + ax + 3 \\
 & -(x^4 - x^3 + x^2) \\
 & \quad 4x^3 - x^2 + ax \\
 & \quad -(4x^3 - 4x^2 + 4x) \\
 & \qquad 3x^2 + (a-4)x + 3 \\
 & \qquad -(3x^2 - 3x + 3) \\
 & \qquad\qquad 0
\end{array}
$$

$$\left(\cancel{3x^2} + (ax-4x) + \cancel{3}\right) - \left(\cancel{3x^2} - 3x + \cancel{3}\right) = 0$$

$$(a-4)x - 3x = 0$$

$$(a-4)\cancel{x} = 3\cancel{x}$$

$$a - 4 = 3$$

$$\therefore a = 1$$

(b.) Hence factorise $h(x)$ Completely

$$h(x) = x^4 + 3x^3 + x + 3$$

$$= (x^2 - x + 1)(x^2 + 4x + 3)$$

$$= (x^2 - x + 1)(x+1)(x+3)$$

$$\therefore h(x) = (x^2 - x + 1)(x+1)(x+3)$$

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6 Solve the inequality

$$3|x-2| > |2x-1|$$

$$|3x-6| > |2x-1|$$

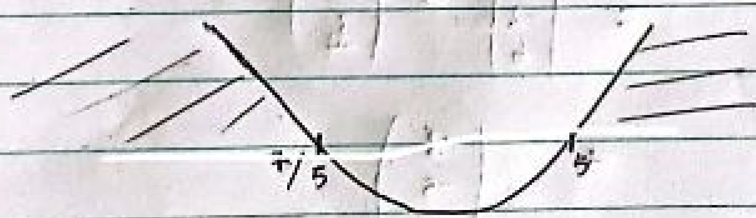
$$(3x-6)^2 > (2x-1)^2$$

$$(3x-6)^2 - (2x-1)^2 > 0$$

$$[(3x-6) - (2x-1)]\,[(3x-6) + (2x-1)] > 0$$

$$(x-5)\,(5x-7) > 0$$

C_v, $x = 5$ or $x = 7/5$



$\therefore\ x < 7/5 \ \cup\ x > 5$

7 (a) $\cos M\hat{O}N = \dfrac{\bar{OM} \cdot \bar{ON}}{|\bar{OM}| \cdot |\bar{ON}|}$

$$= \frac{\begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 6 \end{pmatrix}}{\sqrt{(2)^2+(3)^2+(-4)^2} \cdot \sqrt{(-1)^2+(-2)^2+(6)^2}}$$

$$\cos M\hat{O}N = \frac{-32}{\sqrt{29} \cdot \sqrt{41}}$$

$M\hat{O}N = 158^\circ$ Correct to degree

(b) Unit Vector in the Direction $\overrightarrow{MN}$

$$\overrightarrow{MN} = \overline{ON} - \overline{OM}$$

$$= \begin{pmatrix} -1 \\ -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix}$$

$$= \frac{-3i - 5j + 10k}{\sqrt{(-3)^2 + (-5)^2 + (10)^2}}$$

$$\hat{MN} = \frac{-3i - 5j + 10k}{\sqrt{134}}$$

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8 (a)

$$\frac{dy}{dx} = 4xy^2$$

$$\int dy\,\frac{1}{y^2} = \int 4x\,dx$$

$$\int \frac{1}{y^2}\,dy = \int 4x\,dx$$

$$-\frac{1}{y} = \frac{2x^2 + c}{1}$$

$$\frac{-1}{2x^2+c} = \frac{\cancel{(2x^2+c)}\,y}{\cancel{2x^2+c}}$$

$$\therefore\quad y = \frac{-1}{2x^2 + C}$$

(b) Particular solution of $\frac{dy}{}$, given that $x = 2$ and $y = 4$

$$-\frac{1}{y} = 2x^2 + c$$

$$-\frac{1}{4} = 2(2)^2 + c$$

$$-\frac{1}{4} - 8 = c$$

$$c = -33/4$$

$$-\frac{1}{y} = 2x^2 + c$$

$$-\frac{1}{y} = 2x^2 - \frac{33}{4}$$

$$-\frac{1}{y} = \frac{8x^2 - 33}{4}$$

$$\frac{1}{y} = \frac{33 - 8x^2}{4}$$

$$\frac{4}{33-8x^2} = y\frac{\cancel{(33-8x^2)}}{\cancel{33-8x^2}}$$

$$\therefore\ y = \frac{4}{33-8x^2}$$

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9. $11^n - 4^n$ is a multiple of 7, for all $n \in Z^+$

Proof for $n = 1$

$11^1 - 4^1$

$= 7$

$= 7(1)$

$\therefore$ The Statement is true for $n = 1$

Assume true for $n = k$

$11^k - 4^k = 7R$

$11^k = 7R + 4^k$

Proof for $n = k+1$

$11^{k+1} - 4^{k+1}$

$11^1 \cdot 11^k - 4^1 \cdot 4^k$

$11(7R + 4^k) - 4 \cdot 4^k$

$77R + 11 \cdot 4^k - 4 \cdot 4^k$

$77R + 7 \cdot 4^k$

$7(11R + 4^k)$

$\therefore$ The Statement is true for $n = k+1$

By Mathematical induction, the Statement is true for $n=1$, $n=k$ and $n=k+1$ for all $n \in Z^+$

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Find $\frac{dy}{dx}$

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(a)

$$x^2 + y^2 - 3x^2y + 5y + 2x = 0$$

$$2x + 2y\frac{dy}{dx} - y \cdot 6x - 3x^2\frac{dy}{dx} + 5\frac{dy}{dx} + 2 = 0$$

$$\frac{dy}{dx}\frac{\cancel{(2y - 3x^2 + 5)}}{\cancel{(2y - 3x^2 + 5)}} = \frac{6xy - 2x - 2}{2y - 3x^2 + 5}$$

$$\therefore \frac{dy}{dx} = \frac{6xy - 2x - 2}{2y - 3x^2 + 5}$$

(b) Equation of normal to the Curve @ (1 : 3)

$$\frac{dy}{dx} = \frac{6(1)(3) - 2(1) - 2}{2(3) - 3(1)^2 + 5}$$

$$= \frac{14}{8}$$

$$= \frac{7}{4}$$

$$\frac{dy}{dx}\text{(normal)} = \frac{-4}{7}$$

$$\frac{dy}{dx} = \frac{-4}{7}$$

@ (1 ; 3)

$$y = mx + c$$

$$3 = -\frac{4}{7}(1) + c$$

$$3 + \frac{4}{7} = c$$

$$c = \frac{25}{7}$$

$$7\left[y = \frac{-4x}{7} + \frac{25}{7}\right]$$

$$7y = -4x + 25$$

$$\therefore \quad 7y + 4x = 25$$

$f(x) = x^2 - 6x + 8 \quad , \; 0 \leq x \leq 3$

11 (a) Find $f^{-1}(x)$

$y = x^2 - 6x + 8$
$= x^2 - 6x + (-3)^2 + 8 - (-3)^2$

$= (x-3)^2 + 8 - 9$
$= (x-3)^2 - 1$

$\sqrt{y+1} = \sqrt{(x-3)^2}$

$\sqrt{y+1} = x - 3$

$x = 3 \pm \sqrt{y+1}$

$f^{-1}(x) = 3 \pm \sqrt{x+1}$

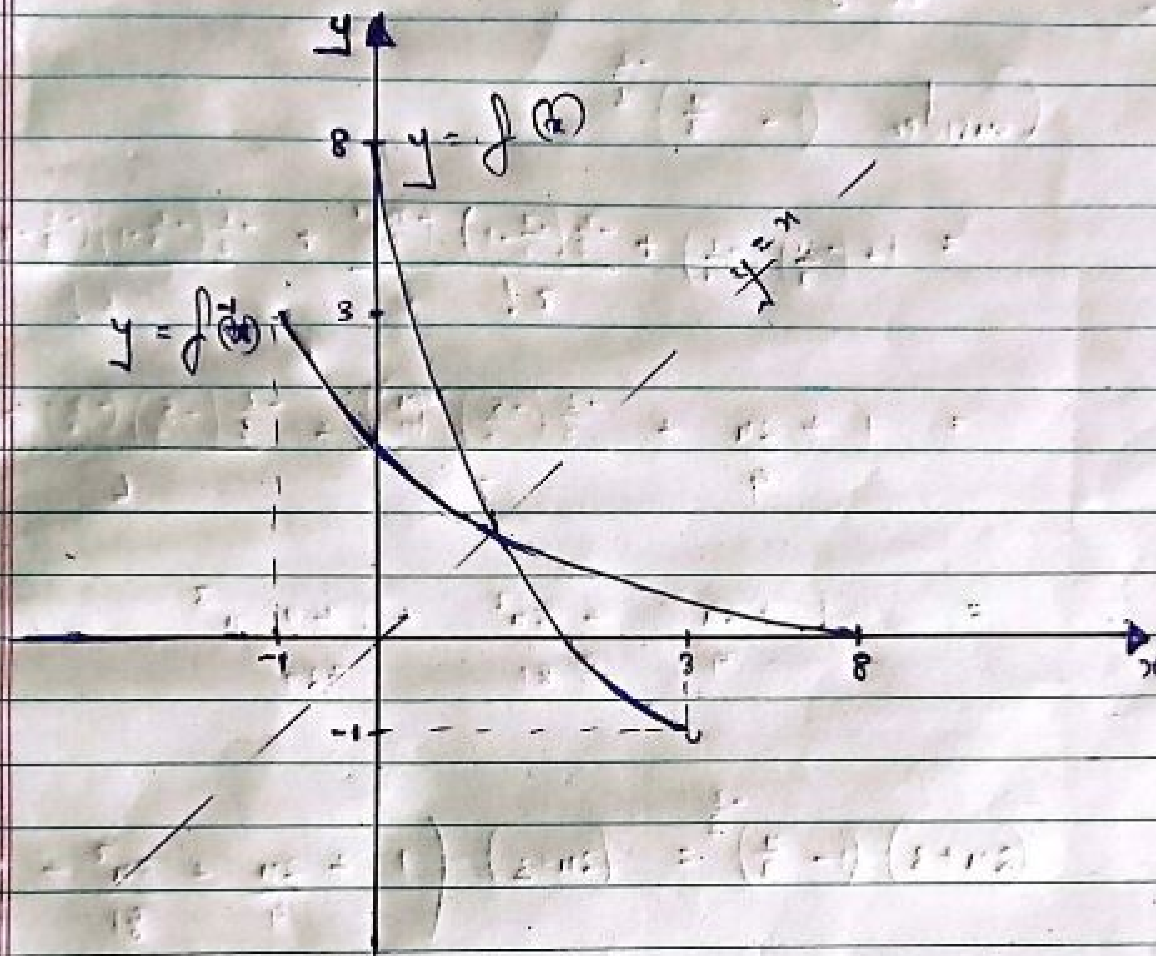
Consider $0 \leq x \leq 3$, take the negative part

$\therefore \; f^{-1}(x) = 3 - \sqrt{x+1}$

(b.) State the domain of $f^{-1}(x)$

$-1 \leq x \leq 8$

(c) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on same axes :



Domain for $f(x)$ :: $0 \leq x \leq 3$

Domain for $f^{-1}(x)$: $-1 \leq x \leq 8$

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12 (a) $\dfrac{2x+3}{\sqrt[3]{\left(1-\frac{x}{3}\right)^2}} = (2x+3)\left(1-\frac{x}{3}\right)^{-\frac{2}{3}}$

Consider $\left(1-\frac{x}{3}\right)^{-\frac{2}{3}}$

$$= 1+\left(-\frac{x}{3}\right)\left(-\frac{2}{3}\right) + \frac{-\frac{2}{3}\left(-\frac{2}{3}-1\right)\left(-\frac{x}{3}\right)^2}{2!} + \frac{-\frac{2}{3}\left(-\frac{2}{3}-1\right)\left(-\frac{2}{3}-2\right)\left(-\frac{x}{3}\right)^3}{3!}$$

$$= 1 + \frac{2x}{9} + \frac{-\frac{2}{3}\left(-\frac{5}{3}\right)\left(\frac{x^2}{9}\right)}{2} + \frac{-\frac{2}{3}\left(-\frac{5}{3}\right)\left(-\frac{8}{3}\right)\left(-\frac{x^3}{27}\right)}{6}$$

$$= 1 + \frac{2x}{9} + \frac{5x^2}{81} + \frac{40x^3}{729}$$

$$(2x+3)\left(1-\frac{x}{3}\right)^{-\frac{2}{3}} = (2x+3)\left(1 + \frac{2x}{9} + \frac{5x^2}{81} + \frac{40x^3}{729}\right)$$

$$= 2x + \frac{4x^2}{9} + \frac{10x^3}{81} + 3 + \frac{6x}{9} + \frac{15x^2}{81} + \frac{120x^3}{729}$$

$$= 3 + \frac{8x}{3} + \frac{17}{27}x^2 + \frac{130}{729}x^3$$

(b) $\left|\frac{x}{3}\right| < 1 \qquad -1 < \frac{x}{3} < 1$

Range

$\therefore \quad -3 < x < 3$

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(a)

$$f(x) = \sqrt{x} + \sqrt{x+1} + \sqrt{x+2} - 5 = 0$$

$$f(1) = \sqrt{1} + \sqrt{2} + \sqrt{3} - 5$$
$$= -0.85$$

$$f(2) = \sqrt{2} + \sqrt{3} + \sqrt{4} - 5$$
$$= 0.14$$

$\therefore$ Since there is sign change, there is a root between 1 and 2

(b)

$$x_0 = 1.5$$

Newton - Raphson - Method

$$f(x) = x^{\frac{1}{2}} + (x+1)^{\frac{1}{2}} + (x+2)^{\frac{1}{2}} - 5$$

$$f'(x) = \frac{1}{2} x^{-\frac{1}{2}} + \frac{1}{2}(x+1)^{-\frac{1}{2}} + \frac{1}{2}(x+2)^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x}} + \frac{1}{2(x+1)^{\frac{1}{2}}} + \frac{1}{2\sqrt{x+2}}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$x_0 = 1.5$

$$x_1 = 1.5 - \frac{f(1.5)}{f'(1.5)}$$

$$= 1.5 - \left[\frac{\sqrt{1.5} + \sqrt{2.5} + \sqrt{3.5} - 5}{\frac{1}{2\sqrt{1.5}} + \frac{1}{2\sqrt{2.5}} + \frac{1}{2\sqrt{3.5}}}\right]$$

$$= 1.825981694$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.838585079$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.838641167$$

$\therefore x = 1.84$ (2d.p)

14 (a) $z^2 + 4z + 7 = 0$

(i)

$z_1 = a + \sqrt{b}i \quad , \quad z_2 = -a - \sqrt{b}i$

$a = 1 \, , \, b = 4 \, , \, c = 7$

$$= \frac{-4 \pm \sqrt{(4)^2 - 4(1)(7)}}{2(1)}$$

$$= \frac{-4 \pm \sqrt{-12}}{2}$$

$$= \frac{-4 \pm \sqrt{4 \times -3}}{2}$$

$$= \frac{-4 \pm 2\sqrt{-3}}{2}$$

$$= \frac{-4}{2} \pm \frac{2\sqrt{3}i}{2}$$

$$= -2 \pm \sqrt{3}i$$

$\therefore \; z_1 = -2 + \sqrt{3}i$ and $z_2 = -2 - \sqrt{3}i$

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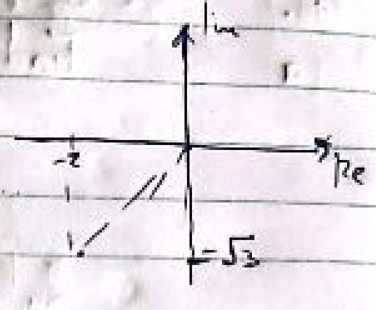
(ii) z_2 in the polar form



$$\therefore z_2 = -2-\sqrt{3}i$$

$$r = \sqrt{(-2)^2+(-\sqrt{3})^2}$$
$$= \sqrt{4+3}$$
$$= \sqrt{7}$$

$$-180 + Tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$
$$= -139$$

$$Tan\,\theta = \left(\frac{-\sqrt{3}}{-2}\right)$$

$$\theta = Tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\theta = -180^\circ + Tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$
$$\theta = -180^\circ + 41^\circ$$
$$\theta = -139^\circ$$

$$z_2 = R(\cos\theta + i\sin\theta)$$

$$\therefore \quad z_2 = \sqrt{7}\left(\cos(-139^\circ) + i\sin(-139^\circ)\right)$$

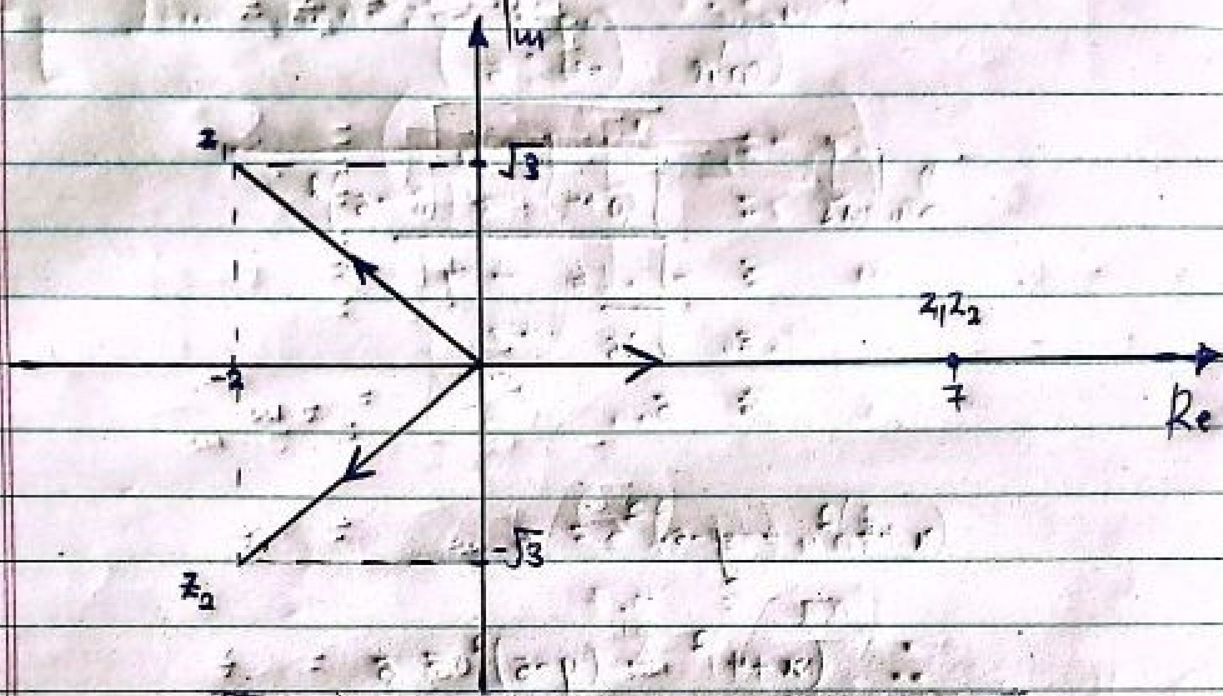
(iii) $z_1 z_2$

$$= (-2+\sqrt{3}i).(-2-\sqrt{3}i)$$

$$= (-2)^2 - (\sqrt{3}i)^2$$
$$= 4 - (-3)$$
$$= 4+3$$
$$= 7$$

$$\therefore z_1 z_2 = 7$$

b.) Represent the Complex numbers z_1 ; z_2 and z_1z_2 on Argand Diagram



15

(a) Find the equation of the Circle

Centre $(-4 ; 5)$

$$\text{Radius} = \sqrt{(0+4)^2 + (8-5)^2}$$
$$= \sqrt{16 + 9}$$
$$= \sqrt{25}$$
$$= 5$$

$$(x+4)^2 + (y-5)^2 = 5$$

$$\therefore \quad (x+4)^2 + (y-5)^2 = 5$$

(b) Show that point B lies on the circumference of the Circle

$$LHS = RHS$$

$$B \; (0 ; 2)$$
$$(x+4)^2 + (y-5)^2 = (5)^2$$

Consider LHS

$$(0+4)^2 + (2-5)^2$$
$$= 16 + 9$$
$$= 25$$

$\therefore$ Since LHS = RHS, the point $(0 : 2)$ lies on the Circumference of the Circle

b. (c) Calculate / Find equation of the Tangent to the Circle at B.

$$\text{Grad (normal)} = \frac{5-2}{-4-0}$$

$$= \frac{-3}{4}$$

$m_1 \times m_2 = -1$

$$\text{Gradient (tangent)} = \frac{4}{3}$$

$y = mx + c$

@ $(0, 2)$

$2 = \frac{4}{3}(0) + c \quad , \quad c = 2$

$\times 3 \quad \left[y = \frac{4}{3}x + 2 \right]$

$\therefore \quad 3y - 4x = 6$

$\therefore \quad 3y - 4x = 6$

d.) Calculate the Coordinates of midpoint of chord BD:

B $(-16 : 8)$; B $(0 : 2)$

$$\left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{-16 + 0}{2} ; \frac{8 + 2}{2} \right)$$

$$= (-8 : 5)$$

e.) Hence or Otherwise, calculate the eqn of perpendicular bisector of BD.

$$\text{Grad} \Rightarrow (BD) = \frac{8 - 2}{-16 - 0} = \frac{-3}{8}$$

$$\text{Grad (perpendicular bisector)} = \frac{8}{3} \rightarrow m_1 \times m_2 = -1$$

$$y = mx + c$$

@ $(-8 ; 5)$

$$5 = \frac{8}{3}(-8) + c$$

$$c = \frac{79}{3}$$

$$\times 3 : \left(y = \frac{8}{3}x + \frac{79}{3} \right)$$

$$\therefore 3y = 8x + 79.$$

16. $y = ab^{-x}$

$$\ln(y) = \ln(a \cdot b^{-x})$$

$$\ln y = \ln a + \ln b^{-x}$$

$$\Rightarrow \ln y = \ln a - x \ln b$$

Plotting the Values $\ln y$ against x

x	1.0	1.5	2.0	2.5	3.0
$\ln y$	1.4	1.7	2.1	2.4	2.8

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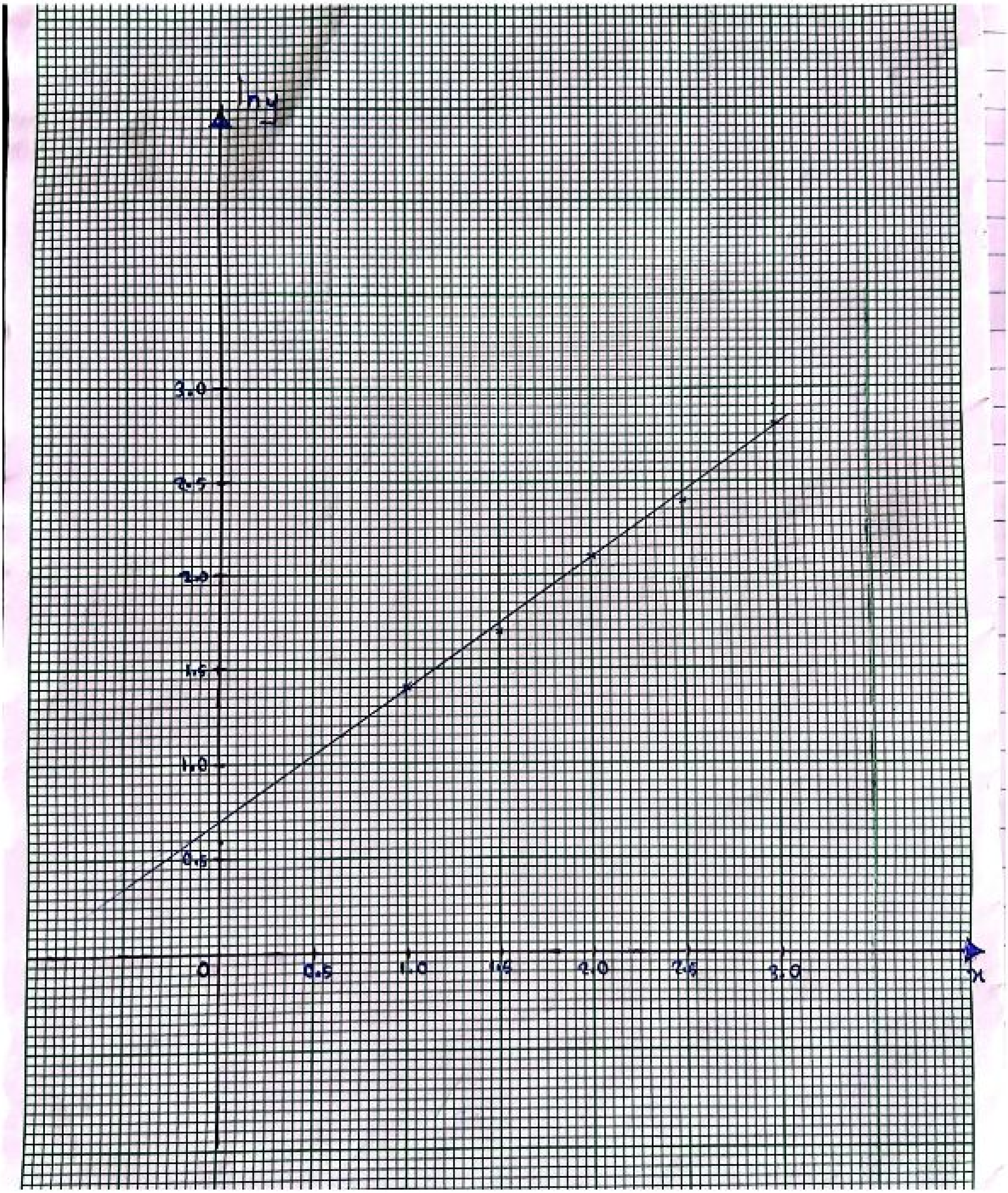

ln y
3.0
2.5
2.0
1.5
1.0
0.5
0
0.5
1.0
1.5
2.0
2.5
3.0
x

6.) Use the graph to find approximate values of a and b

Compare $\ln y = \ln a - x\ln b$ with $y = mx + c$

$\ln y : y$

$m = -\ln b$

$c = \ln a$

$\ln a = c$

$e^{\ln a} = e^{0.7}$

$a = e^{0.7}$

$a = 2.013\ 752\ 707$

$-\ln b = \left(\frac{2.8 - 1.4}{3 - 1}\right)$

$-\ln b = \frac{7}{10}$

$-\ln b = 0.7$

$e^{\ln b} = e^{-0.7}$

$b = e^{-0.7}$

$= 0.496\ 585\ 303$

$\therefore$ $a = 2.0$ and $b = 0.50$ (2 s.f)